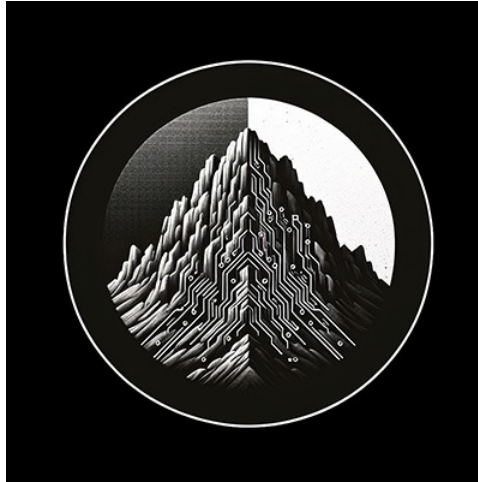




RTL Design Sherpa

Pre-Synth Timing Prediction A White Paper on Designing-for-Close Before Synthesis Ever Starts

June 12, 2026

Table of Contents

1 Readme.....	6
2 A Portable, Macro-Optional ASIC Timing Characterization Harness.....	6
2.1 Abstract.....	6
2.2 1. Why this harness, and not just RTL we already have?.....	7
2.3 2. High-level architecture.....	9
2.4 3. The 3-edge pipeline.....	9
2.5 4. Anti-optimisation strategy.....	10
2.6 5. The FUB catalog.....	11
2.7 6. The two source trees.....	12
2.7.1 6.1 rtl/ – the macro-driven default.....	13
2.7.2 6.2 rtl/asic_only/ – the macro-free ASIC fork.....	13
2.8 7. The synthesis flow.....	14
2.8.1 7.1 Open-source flow (Yosys + OpenSTA, asic_only tree).....	14
2.8.2 7.2 Commercial reference flows (asic_only tree).....	15
2.9 8. Reading the reports.....	15
2.9.1 8.1 Comparing technologies.....	16
2.10 9. Verification.....	17
2.11 10. Directory layout.....	17
2.11.1 10.1 Regenerating the diagrams.....	18
2.12 11. Limitations and intentional non-features.....	19
2.13 13. From characterization to design: the ASAP7 worked example.....	19
2.13.1 13.1 Sheet 1: characterization – the technology data.....	20

2.13.2 13.2 Sheet 2: building blocks – per-block SWAGs.....	21
2.13.3 13.3 Sheet 3: stream – per-FUB port trace.....	22
2.13.4 13.4 The workflow: planning at the spreadsheet.....	23
2.13.5 13.5 Grounding: how the data got there.....	24
2.13.6 13.6 The point.....	24
2.14 14. Navigation.....	25

List of Figures

No figures in this document.

List of Tables

No tables in this document.

1 Readme

Generated: 2026-06-12

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2 A Portable, Macro-Optional ASIC Timing Characterization Harness

A practitioner's guide to `char_top` – a benchmarking harness that measures combinational delay across ASIC standard-cell libraries by forcing the synthesis tool to build the structures we actually want it to build.

Version: 3.1 (ASAP7 worked example landed) **Last Updated:** 2026-06-12 **Status:** RTL + verification + ASIC synthesis flows in place; ASAP7 RVT characterization run + budget worksheet live under `work/`. (45 tests passing; Yosys + OpenSTA + Synopsys DC + Cadence Genus drivers; 99-row ASAP7 sweep at TT/FF/SS clean.)

An FPGA-side companion – with the same workflow and an analogous spreadsheet – is published as a separate addendum README. This document is ASIC-only.

2.1 Abstract

The sole purpose of this methodology is to allow the designer to predict and close timing **before synthesis ever starts**. This flow lets the designer characterize various structures at various technology nodes and frequencies so that, by the time the production design enters synthesis, each combinational shape has a known per-node, per-frequency budget – and the answer to “will this pipeline stage close?” is already on a table. The same characterization run also yields an **accurate per-structure gate count**: because the harness pins every internal net through `dont_touch` and emits a per-FUB cell-count + area summary from both Yosys (`build/$TOP.stat.txt`) and the commercial flows (`report_area`), the designer gets a real, technology-mapped gate-count estimate for any combinational shape long before the production netlist exists.

This methodology is **explicitly proactive**. It is not a debugger you reach for after a synthesis run reveals a violation; it is a design-time budget the designer (or uArch) consults *before writing the first `always_ff`*. The whole point is to flip the loop – characterize once per node, then build with a known-good budget – so synthesis becomes a confirmation step rather than the first time anybody finds out the pipeline doesn’t close. Concretely, the methodology exists **to prevent, as much as possible, the designer being surprised when synthesis happens**: every shape on the critical path has already been measured at this node and frequency, every gate count has already been totalled, and the answer is on the worksheet long before the synthesis tool ever weighs in.

`char_top` is a synthesizable harness that wraps nine **Functional Unit Blocks (FUBs)** – each a different shape of combinational logic – between registered endpoints driven by a shared 32-bit Galois LFSR. By driving every flop input with a maximal-length pseudo-random sequence and routing every flop output to a top-level port, the design forecloses every optimisation a modern synthesis tool would otherwise use to make the benchmark disappear (constant propagation, dead-code elimination, common-subexpression elimination, retiming-into-pads, and so on). Within that frozen scaffold, each FUB exposes a specific combinational pattern (NAND tree, ripple-carry adder, mux tree, structural multiplier, ...) so that the slack reported for its path group is a clean, comparable measure of that pattern’s depth in the target technology.

The component ships in two source-equivalent flavours, both targeted at ASIC characterization:

- **rtl/** – the default, macro-driven tree that supports compile-time switches for reset style (`USE_ASYNC_RESET`, `RESET_ACTIVE_HIGH`) and FIFO memory type. Use this tree when the cell library has multiple flop topologies (sync / async reset, active-low / active-high) and you want one source to characterise all of them.
 - **rtl/asic_only/** – a macro-free fork hard-wired to async-negedge active-low reset. Use this tree when the library has exactly one flop topology (async, active-low) and you want the source you read to match the gates that get built, with no preprocessor in the way. Ships with an open-source Yosys + OpenSTA flow plus reference scripts for Synopsys Design Compiler and Cadence Genus.
-

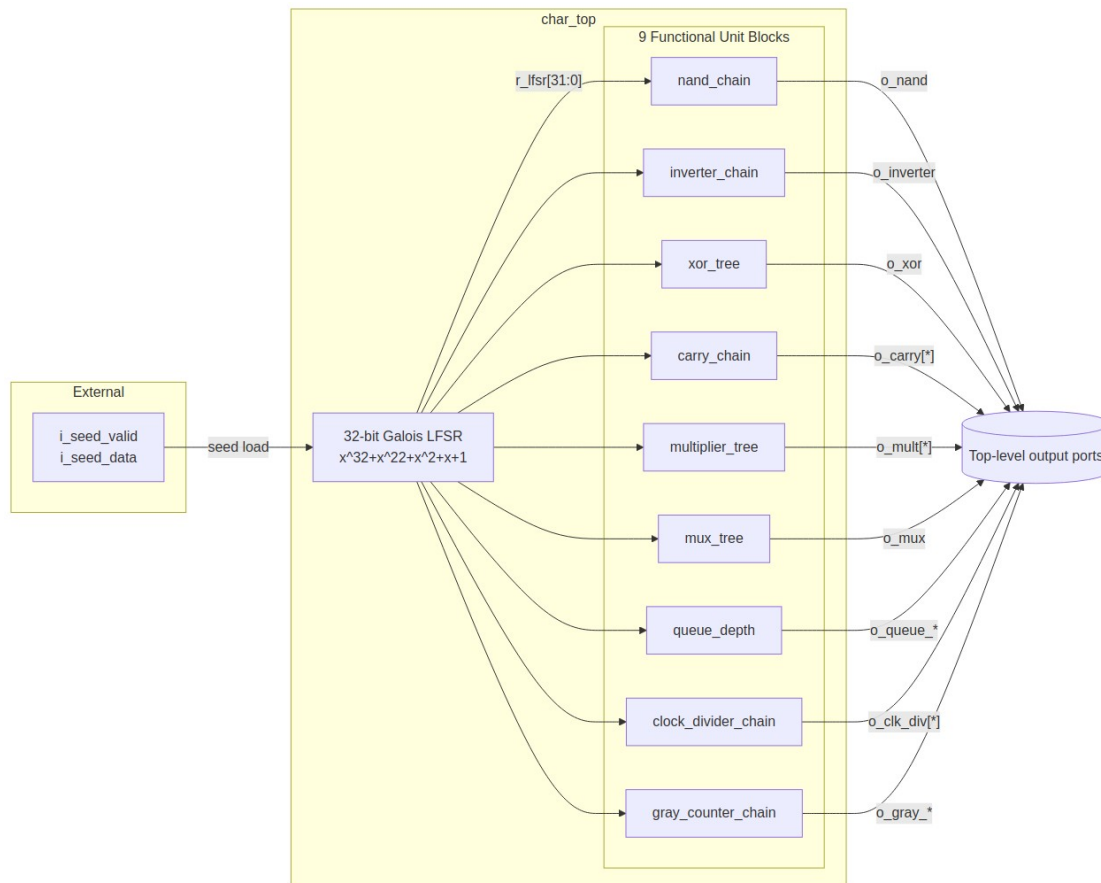
2.2 1. Why this harness, and not just RTL we already have?

A synthesis tool is, by design, free to delete logic whose output is unobserved or whose input is constant. Without explicit countermeasures any benchmark “design” collapses to a few wires before it ever reaches mapping. This harness solves four problems at once:

1. it gives every standard combinational pattern (tree, chain, multiplier, mux, memory, counter) a **registered envelope** so STA reports the pattern's own delay, not an I/O delay;
2. it drives every flop input from a **shared maximal-length LFSR** so the tool cannot prove any input constant;
3. it routes every flop output to a **top-level port** so the tool cannot prune any output;
4. it carries (* dont_touch = "true" *) on every chain net as a belt-and-suspenders defence against aggressive remapping.

The result is a single source that runs through any flow you point at it and lets you measure – not estimate – how a 12-level XOR tree, or a 64-bit ripple-carry adder, or a 16x16 Dadda multiplier behaves in the target technology.

2.3 2. High-level architecture

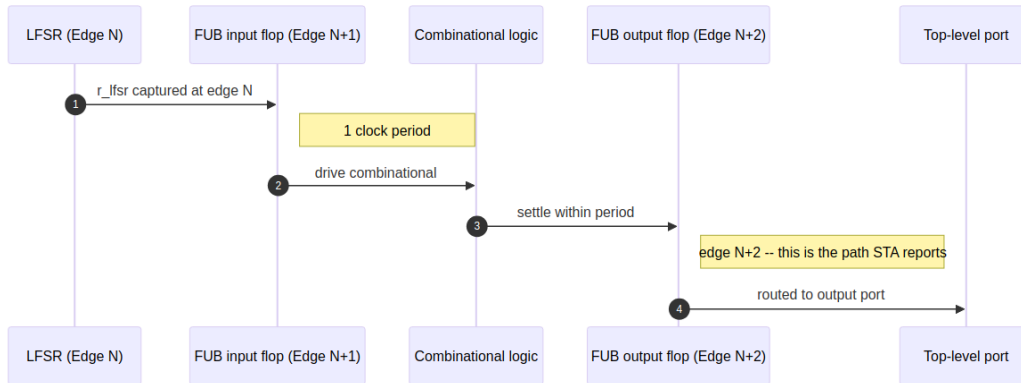


Architecture overview

Each FUB is gated by an `EN_*` parameter. When disabled (`EN_*=0`) the corresponding generate block emits a tie-to-zero and consumes zero logic – this is how targeted “just the multiplier” or “everything but the FIFO” sweeps work.

2.4 3. The 3-edge pipeline

The harness deliberately registers both endpoints of every FUB’s combinational logic so STA reports slack for the register-to-register path itself, not the I/O path. Because the LFSR is the source for every input, the design has a fixed three-edge pipeline:

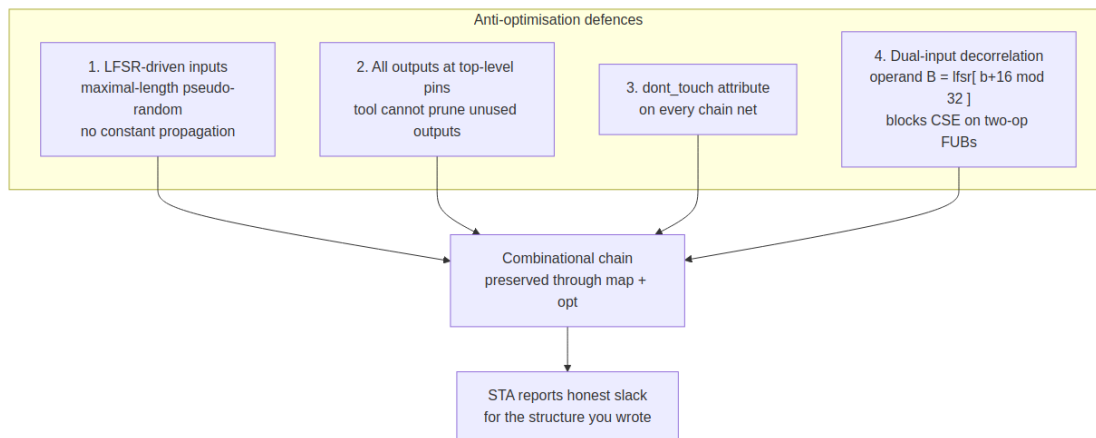


3-edge pipeline

A `char_top`-level testbench that compares an expected value against an observed output sampled after rising edge C reads `lfsr_idx = C - 2` from the reference model (`dv/tbclasses/timing_char_tb.py:lfsr_step / lfsr_sequence`).

The LFSR in `char_top` is a **left-shift** Galois LFSR (polynomial `0x0040_0007`). This is architecturally different from the right-shift tap-based Galois LFSR in `val/common/test_shifter_lfsr_galois.py` – do **not** reuse that model.

2.5 4. Anti-optimisation strategy



Anti-optimisation defences

Defence	Threat it blocks
LFSR-driven inputs	Constant propagation, dead-input elimination
Outputs at the pads	Dead-output elimination, sweep

Defence	Threat it blocks
dont_touch attribute	Aggressive remapping that erases internal nets
Dual-input decorrelation (r_lfsr[(b+16) % 32])	CSE on a OP a

All four work together. Remove any one and a sufficiently aggressive synthesis tool will find a shortcut; with all four in place the gates you wrote are the gates that land in the netlist.

2.6 5. The FUB catalog

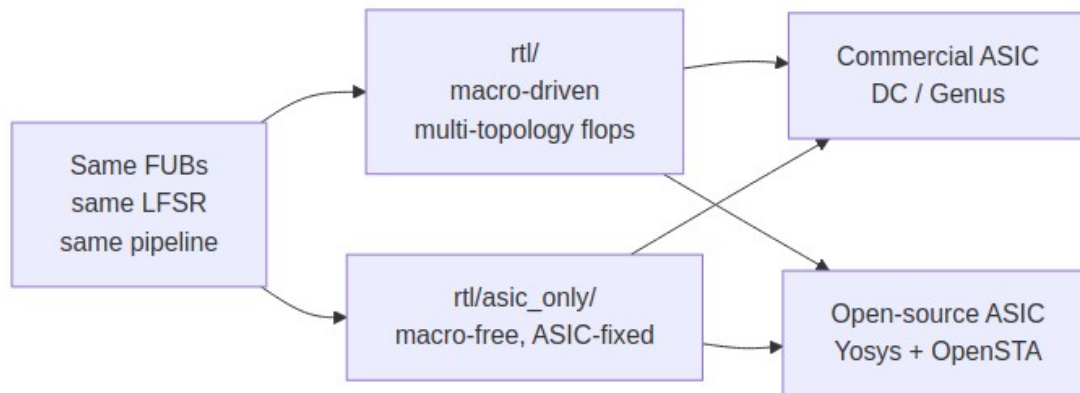
FUB	File	Key parameters	Critical path	What it measures
nand_chain	fub/ nand_chain.sv	LEVELS, NUM_FLOPS	LEVELS NAND-gate delays	Raw 2-input NAND propagation through the target technology
inverter_chain	fub/ inverter_chain.sv	INVERTER_COUNT	Linear inverter chain	Best-case per-gate delay (no fan-in overhead)
xor_tree	fub/ xor_tree.sv	LEVELS, NUM_FLOPS	LEVELS XOR delays	XOR cell vs. NAND cell / LUT packing
carry_chain	fub/ carry_chain.sv	WIDTH	WIDTH carry-propagation delays	Dedicated carry-chain (CARRY4 / CARRY8 / ALM-carry / stdcell ripple)
multiplier_tree	fub/ multiplier_tree.sv	WIDTH, MULT_TYPE	Reduction-tree depth	Inferred * (DSP) vs. structural

FUB	File	Key parameters	Critical path	What it measures
				Dadda / Wallace / CSA
mux_tree	fub/ mux_tree.sv	LEVELS, NUM_FLOPS	LEVELS 2:1- mux delays	Mux-cell delay vs. logic-cell delay
queue_depth	fub/ queue_depth. sv	DATA_WIDTH, DEPTH	FIFO read path + output flop	Memory inference thresholds (LUTRAM / BRAM / URAM / SRAM compiler)
clock_divide r_chain	fub/ clock_divide r_chain.sv	STAGES, CW, PICKOFF	Counter increment + pickoff mux	Counter- based clock generation
gray_counter _chain	fub/ gray_counter _chain.sv	WIDTH	Binary carry + XOR tree	Binary-to- Gray conversion delay

Per-FUB rationale, recommended sweep ranges, and result interpretation live in `rtl/syn/SYNTHESIS_GUIDE.md`.

2.7 6. The two source trees

The component ships two source-equivalent trees so that the same nine FUBs can be characterised with the right amount of preprocessor indirection for the job.



Two source trees

2.7.1 6.1 rtl/ - the macro-driven default

The default tree includes `reset_defs.svh` and `fifo_defs.svh` so a single source set covers every flop topology a target ASIC library might expose:

Switch	Effect
<code>\USE_ASYNC_RESET</code>	Includenegedge
<code>rstin</code>	the flop sensitivity list
<code> `RESET_ACTIVE_HIGH</code>	Flip reset polarity
<code> MEM_STYLE =</code>	
<code>FIFO_{AUTO,SRL,BRAM}</code>	Drive the inferred memory style
<code>ofgaxi_fifo_sync`</code>	

Use this tree when the cell library you’re characterising has more than one flop topology (sync-reset cells in addition to async, or active-high reset cells in addition to active-low) and you want a single source sweep all of them.

2.7.2 6.2 rtl/asic_only/ - the macro-free ASIC fork

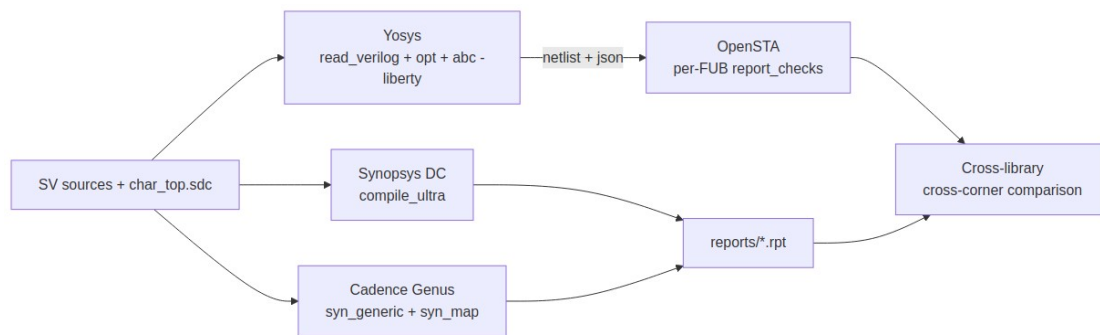
The `asic_only` fork strips all three abstractions:

- `\include “reset_defs.svh”` and `\include “fifo_defs.svh”` removed
- `\ALWAYS_FF_RST(clk, rst, BODY)` rewritten to `always_ff @(posedge clk or negedge rst) begin BODY end``
- `\RST_ASSERTED(rst)` rewritten to `!(rst)``
- `fifo_mem_t` parameter type replaced with `int`; `FIFO_AUTO`, `FIFO_SRL`, `FIFO_BRAM` inlined as 0, 1, 2

The behaviour is **identical** to the upstream tree when configured for `USE_ASYNC_RESET` + active-low; the `asic_only` variant just bakes that single posture into source so every flop is one obvious `always_ff` block and every reset condition is one `!rst_n` test. Use this tree when:

- the target library exposes a single flop topology (async, active-low) anyway, so the macro switches are dead options;
- you want the source you read to match the netlist the tool builds, with no preprocessor gates in between;
- you want the open-source Yosys + OpenSTA flow without library-defs in your include path.

2.8 7. The synthesis flow



Synthesis flow back ends

Three ASIC back ends are wired up: one open-source (Yosys + OpenSTA), two commercial (Synopsys DC, Cadence Genus). All three source the same `char_top.sdc` so reports use the same path groups and are directly comparable.

2.8.1 7.1 Open-source flow (Yosys + OpenSTA, asic_only tree)

```
cd projects/components/timing_characterization/rtl/asic_only/syn
```

```
# Minimal invocation -- 500 MHz, every FUB enabled
make report LIB_PATH=/path/to/stdcell.lib
```

```
# 800 MHz, no multiplier (isolates the rest of the harness)
make report LIB_PATH=/path/to/stdcell.lib \
    TARGET_FREQ_MHZ=800 EN_MULTIPLIER=0
```

```
# Just rerun STA against the existing netlist
make sta LIB_PATH=/path/to/stdcell.lib
```

Outputs land under syn/build/:

File	Contents
build/\$TOP.netlist.v	Post-mapped gate-level netlist
build/\$TOP.json	Yosys JSON (for graph tools, formal, ABC reruns)
build/\$TOP.stat.txt	Cell-count + area summary per Liberty
build/sta.log	Full OpenSTA log incl. WNS per group
build/timing/GRP_*.rpt	Per-FUB worst-path report
build/area.rpt	Hierarchical area
build/yosys.log	Synthesis trace for diffing across runs

2.8.2 7.2 Commercial reference flows (asic_only tree)

```
dc_shell -f rtl/asic_only/syn/dc_synth.tcl          # Synopsys DC
genus    -files rtl/asic_only/syn/genus_synth.tcl  # Cadence Genus
```

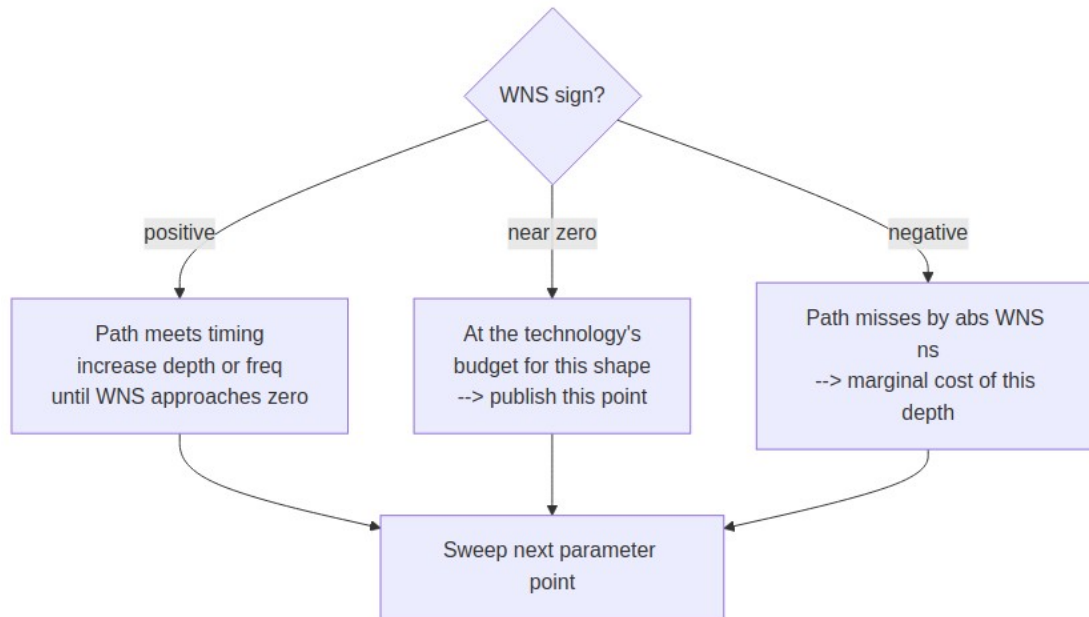
Both reference scripts source the same char_top.sdc and declare the same per-FUB path groups, so reports are directly comparable to the open-source flow's output.

2.9 8. Reading the reports

The interesting line in the OpenSTA log (or the DC / Genus equivalent) is the per-group worst-negative-slack:

```
Path Group: GRP_NAND      WNS = 0.1234 ns
Path Group: GRP_INVERTER  WNS = 0.4567 ns
...
```

Positive WNS = the FUB meets the target frequency with that much slack to spare. Negative WNS = the FUB misses by that much; the absolute value is the additional period the technology needs at this depth.



WNS-sign decision tree

Sweep protocol:

1. Fix TARGET_FREQ_MHZ.
2. For each FUB, sweep its depth / width parameter until WNS crosses zero.
3. Record the parameter value at WNS = 0 – that's the technology's max depth for that pattern at the chosen frequency.
4. Repeat across libraries / devices for an apples-to-apples cross-target view.

2.9.1 8.1 Comparing technologies

Run the same sweep against two libraries (asic_only flow):

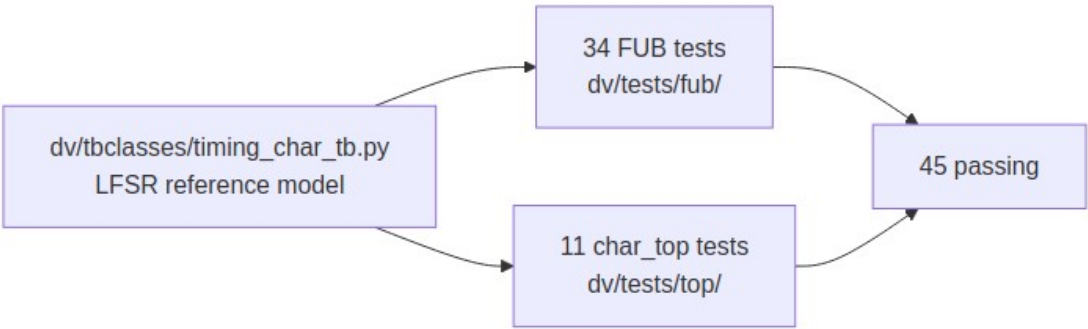
```
make report LIB_PATH=lib_A.lib TARGET_FREQ_MHZ=500
mv build/timing build/timing.A
```

```
make clean
make report LIB_PATH=lib_B.lib TARGET_FREQ_MHZ=500
mv build/timing build/timing.B
```

```
diff -u build/timing.A/GRP_CARRY.rpt build/timing.B/GRP_CARRY.rpt
```

The path structures are identical; only the cell delays differ. Any WNS delta therefore comes purely from the technology, not from a synthesizer mood swing or a netlist topology change.

2.10 9. Verification



Verification topology

Suite	Location	Tests	Status
Per-FUB sweeps	dv/tests/fub/	34	All passing
char_top enable-mix configurations	dv/tests/top/	11	All passing

```
PYTHONPATH=bin:$PYTHONPATH pytest \
  projects/components/timing_characterization/dv/tests/ -v
```

The TB class lives in `dv/tbclasses/timing_char_tb.py` and implements the mandatory three methods (`setup_clocks_and_reset`, `assert_reset`, `deassert_reset`); the `lfsr_step / lfsr_sequence` reference functions match the left-shift LFSR baked into `char_top`.

2.11 10. Directory layout

```
projects/components/timing_characterization/
├── README.md          <-- this file
├── CLAUDE.md          AI assistance guide
├── PRD.md             Product Requirements
├── TASKS.md           Task tracking
├── assets/            diagram sources + rendered PNGs
│   ├── Makefile      `make` to re-render every *.mmd ->
│   └── *.png
├── puppeteer.json     mmdc no-sandbox config
├── *.mmd              diagram source (mermaid)
├── *.png              diagram renders (embedded above)
├── docs/              Characterization results (future)
├── dv/
│   └── tbclasses/
│       └── timing_char_tb.py  TB base + LFSR reference model
```

```

├── tests/
│   ├── fub/          9 FUB tests (34 parameter points)
│   └── top/          char_top tests (11 enable-mixes)
├── work/             <-- the worked example artifacts
│   ├── asap7_characterization.xlsx 3-sheet methodology workbook
│   ├── timing_char_sweep.py       9-primitive 2-point probe driver
│   └── timing_char_data.csv        54-row primitive characterization
├── data
│   ├── abc_sweep.py              11-row STREAM library sweep driver
│   ├── timings.csv               99-row per-block ASAP7 sweep output
│   ├── build_characterization.xlsx CSV -> xlsx generator
│   └── merge_lib_corner.py       ASAP7 RVT Liberty merger (TT/FF/SS)
├── rtl/
│   └── common/                  shared primitives (counters, math,
gaxi_fifo_sync)
│   ├── fub/                    9 FUBs
│   └── filelists/char_top.f     flat source list for the multi-flow
├── tree
│   ├── top/                    char_top.sv + nand_chain_top.sv
│   ├── syn/                    multi-flow SDC + SYNTHESIS_GUIDE
│   └── asic_only/              <-- macro-free ASIC fork
│       ├── common/             same primitives, macros expanded
│       ├── fub/                 same FUBs, macros expanded
│       ├── filelists/char_top.f flat source list (asic_only)
│       └── top/                 char_top.sv + nand_chain_top.sv
├── (asic_only)
│   └── syn/                     ASIC-only flow
│       ├── Makefile             Yosys + OpenSTA driver
│       ├── yosys_synth.py
│       ├── sta_run.tcl
│       ├── dc_synth.tcl         Synopsys DC reference
│       ├── genus_synth.tcl      Cadence Genus reference
│       ├── char_top.sdc         ASIC-only SDC
│       └── SYNTHESIS_GUIDE.md

```

2.11.1 10.1 Regenerating the diagrams

The seven figures in this document live as mermaid source in assets/ and are rendered to PNG by mmdc (@mermaid-js/mermaid-cli). Re-render after editing the source:

```

cd projects/components/timing_characterization/assets
make                # render every *.mmd -> *.png
make clean          # delete the PNGs

```

assets/puppeteer.json passes --no-sandbox so headless Chrome runs inside sandboxed environments and CI runners.

2.12 11. Limitations and intentional non-features

- **No memory compiler integration.** `queue_depth` is mapped by the tool to whatever the Liberty file exposes; an SRAM-compiler wrapper is out of scope.
 - **No SDF back-annotation.** STA against the synth Liberty is wire-load estimated, not parasitic-aware. Round-trip P&R + extraction is out of scope of a *characterization* harness.
 - **No power numbers from the open-source flow.** Add a switching activity file and a power-aware Liberty to extend the OpenSTA invocation with `report_power`.
 - **asic_only is async-reset only.** The harness models the worst case (async reset path through every flop) so the cell library's reset recovery / removal delay is reflected in the report. Sync-reset numbers come from the macro-driven rtl/ tree with `USE_ASYNC_RESET` left undefined.
-

2.13 13. From characterization to design: the ASAP7 worked example

The methodology's design-time artifact is a single workbook with three linked sheets: `work/asap7_characterization.xlsx`. It's built bottom-up against ASAP7 RVT (the same flow described in section 7) and lets a designer answer "will this pipeline close?" by changing one cell – before any RTL exists for the production block.

The three sheets stack:

1. **characterization** – the technology data. Per-gate-type delay (NAND, XOR, MUX, CARRY, MULT, ...) at TT / FF / SS, plus Tflop, wire delay per metal layer, and the **Design clocks table** that names the clocks the design uses.
2. **building blocks** – a SWAG estimator for every reusable RTL block (skid buffer, `gaxi_fifo_sync`, `arbiter_round_robin`, `axi4_master_rd/wr`, SRAMs). Each row's `data->D` / `flop->out` / `fmax` / `slack` are formulas that look up the planning corner and freq from the characterization sheet.
3. **stream** – a worked example: every port of every STREAM FUB walked by hand, one row per (FUB, port) pair. The slack cell on each row is corner- and freq-aware via VLOOKUP into the clocks table.

Editing the clocks table on sheet 1 propagates to slack columns on sheets 2 and 3. Negative slack cells render **bold red** so failing paths jump off the page during the sweep.

2.13.1 13.1 Sheet 1: characterization – the technology data

The top of the sheet is a sweep grid: TT / FF / SS x five frequencies (currently 2.0 / 2.25 / 2.5 / 2.75 / 3.0 GHz). Each cell shows what's achievable in that corner+freq combination. The two driving rows are:

- **Per-level gate delay** (ps, freq-independent) – the slope from a two-point synth probe per primitive at that corner. This is what gets pulled by the downstream sheets.
- **Max combinational levels per cycle** – the budget: $(\text{period} - T_{\text{flop}}) / \text{per_level_delay}$. Tells you how deep a chain of that gate type can be before it busts the period.

TT corner, freq sweep	NAND	XOR	MUX	CARRY	MULT (proxy)	GRAY_CTR	QUEUE
per-level delay (ps)	7.25	26.8	16.3	30.8	30.0	58.4	11.6
max levels @ 2.0 GHz	66	18	29	15	16	8	41
max levels @ 3.0 GHz	43	12	19	10	10	5	27

The **MULT** row is highlighted yellow on the sheet – it doubles as the recommended proxy for mixed-bag combinational logic (a real mult tree contains AND, XOR, full-adder, and carry cells, so its per-level delay blends realistically).

Below the per-corner data lives the **Design clocks table** – one configurable row per clock the design uses (up to six slots):

clock name	freq (MHz)	target corner	period (ps)	input 60%	output 40%	wire/hop
clk_main (5 empty slots)	1000	TT	=1e6/B	=D*0.6	=D*0.4	100

Yellow cells are editable; the rest derive. This table is what sheets 2 and 3 read – their slack columns are wired here via VLOOKUP.

Last two sections on the sheet: **Wire delay per mm by metal layer** (M2/M3 through M8/M9 – 120 to 22 ps/mm TT), and notes.

2.13.2 13.2 Sheet 2: building blocks – per-block SWAGs

One row per reusable RTL block, in two sections. The **SRAMs** section holds the storage primitives:

block	params	flops	NAND_eq	data->D (TT, ps)	flop->out	slack @ clk_main
SRAM (raw macro)	W=512, D=512	0	~2110	7	7	(depends on clock)
gaxi_fifo_sync (REG=1, SRAM- backed)	W=512, D=512	28	2188	7	7	...

flops on the SRAM rows is the **control-only** count – storage bits ($W * D$) live in the BRAM/SRAM macro, not in std cells. A reference block below the table shows storage = $W*D$ in bits and KB.

The **Building Blocks** section follows with the rest:

block	base params	flops	NAND_eq	in mux/nand	out mux/nand	slack @ clk_main
gaxi_skid_buffer	W=1, D=2	6	18	1 / 1	0 / 0	(compute d)
gaxi_fifo_sync (REG=0)	W=1, D=2	6	40	1 / 2	$\log_2(D) / 0$	...
arbiter_round_robin	N=2	4	16	0 / 1	$\log_2(N) / 1$	...
axi4_master_rd	AW=DW=32, IW=4	212	848	1 / 1	1 / 1	...
axi4_master_wr	similar	222	888	1 / 1	1 / 1	...

Each row's clock column (yellow) defaults to clk_main – change it to point a block at a different clock entry on sheet 1. data->D covers the input-side combinational delay; flop->out covers the output-side; delay_in defaults to 30 percent of the target period (STA input arrival); slack = period - delay_out - 40 percent of period - wire_per_hop.

Skid buffers have `out_mux = out_nand = 0` because the head flop's Q drives the output directly – there's no internal output muxing. The other blocks have non-zero output-side levels (the read mux tree on FIFOs, the priority encoder on the arbiter, the skid output select on the AXI masters).

2.13.3 13.3 Sheet 3: stream – per-FUB port trace

This is the worked example proper: 11 STREAM FUBs, 144 rows total, one row per (FUB, port) pair. Walking `rtl/fub/stream_latency_bridge.sv` by hand gave the first ten rows; the other ten FUBs were filled in the same way.

Each row has:

col	content
A	FUB name
B	port name
C	clock (defaults <code>clk_main</code> ; edit to point at a different clock)
D	direction (in / out / local)
E	building block crossed (if the port enters / leaves a BB)
F	MUX lv on this row's path
G	NAND lv on this row's path
H	source/sink flop or BB port the path terminates at
I	target period (VLOOKUP from clocks table)
J	target corner (VLOOKUP)
K	local delay (corner-aware: $F * \text{per_MUX}(\text{corner}) + G * \text{per_NAND}(\text{corner})$)
L	<code>delay_in</code> (30 percent of target period by default; override with <code>=M_prev</code> to chain)
M	<code>delay_out = delay_in + local delay</code>
O	<code>slack = period - delay_out - output_40% - wire/hop</code>
P	places it goes (which downstream

col	content
	FUB / CSR / system bus consumes this)

A local direction marks an internal flop hop (r_state -> r_next inside one FUB). 39 of the 144 rows are local-flop hops; 26 cross a building block boundary; the rest are external ports.

The “places it goes” column carries the partition story. For example:

FUB	port	dir	terminator	places it goes
stream_latency_bridge	m_valid	out	u_skid_buffer.r.rd_valid (Q)	to sram_controller_unit.axi_wr_sram_val id -> axi_write_engine
scheduler	axi_rd_req_val	out	r_axi_rd_req (Q)	to axi_read_engine (per-channel request)
descriptor_engine	descriptor_val	out	i_descriptor_fifo.rd_val id (Q)	to scheduler (descriptor dispatch)

2.13.4 13.4 The workflow: planning at the spreadsheet

The whole point of the three-sheet stack is to make pipeline planning a spreadsheet exercise:

1. Open the workbook, go to characterization, find the Design clocks table, set clk_main freq + corner to your target.
2. Flip to building blocks – which BB rows light up red in the slack column? Those are the blocks you need to either pipeline harder or stop relying on at this freq.
3. Flip to stream – which FUB ports are red? Those tell you which inter-FUB hops won’t close. The “places it goes” column maps a red row back to a concrete RTL location.
4. Iterate: lower the freq until everything’s green (the design’s actual ceiling); switch to FF corner to model best-case silicon; bump wire/hop to model long routes.

Three knobs to play with:

- **clk_main freq** (B29) – walk the period; watch slack flip.
- **clk_main target corner** (C29) – TT for sign-off, FF for best-case silicon, SS for worst-case.
- **clk_main wire/hop** (G29) – default 100 ps; bump to 200-300 for long inter-FUB routes, drop to 0 for paths that stay inside one FUB.

Add more clocks (clk_apb, clk_cdc, ...) in slots 30-34 and re-point specific rows on sheets 2 / 3 by editing their clock cell.

2.13.5 13.5 Grounding: how the data got there

The numbers on sheet 1 come from a single Python driver `work/timing_char_sweep.py` that:

1. Reads the nine `asic_only/fub/*.sv` primitives.
2. Synthesizes each at two probe sizes (small + large) per corner through Yosys + ABC against ASAP7 RVT, using `work/merge_lib_corner.py`'s merged Liberty.
3. Reads ABC's `stime -c` output for each run to get delay (ps) and logic-level count.
4. Derives per-level slope and T_{flop} via linear extrapolation, drops them in `work/timing_char_data.csv`.

Then `work/build_characterization_xlsx.py` turns the CSV plus hand-walked FUB tables into the three-sheet workbook.

A parallel driver `work/abc_sweep.py` runs every STREAM library row (11 distinct blocks) through the same Yosys + ABC flow at TT / FF / SS x three target frequencies (99 cells, all clean, ~9 minutes wall-clock). Output is `work/timings.csv`; it's the "ground truth" measurement to calibrate the SWAGs on sheet 2 against – the SWAG model is first-order and underestimates real wide-bus buffering by ~10x, so production planning should treat the spreadsheet slack as an upper bound and lean on the CSV for the hard numbers.

2.13.6 13.6 The point

The same exercise can run against any ASIC technology you point the flow at: rerun the per-primitive probes against a different Liberty, regenerate the workbook, and the methodology recomputes. The characterization runs **once per (technology, corner)**; the rest of the workflow is a spreadsheet.

The same workflow runs on an FPGA target – different delay model, different primitives, different gate units – and the worksheet shape stays

the same; only the library columns change. That companion is the subject of a separate addendum README, not this document.

2.14 14. Navigation

- AI assistance guide: `CLAUDE.md`
- Product requirements: `PRD.md`
- Task tracking: `TASKS.md`
- Per-FUB synthesis recipes: `rtl/syn/SYNTHESIS_GUIDE.md`
- Multi-flow constraint file: `rtl/syn/char_top.sdc`
- ASIC-only constraint file: `rtl/asic_only/syn/char_top.sdc`
- Open-source flow driver: `rtl/asic_only/syn/Makefile`
- Design-time budget worksheet: `work/asap7_characterization.xlsx`
- Workbook generator: `work/build_characterization.xlsx.py`
- Per-primitive ASAP7 probe driver: `work/timing_char_sweep.py`
- Per-block STREAM library sweep: `work/abc_sweep.py`
- Merged ASAP7 RVT Liberty builder: `work/merge_lib_corner.py`
- Primitive characterization data: `work/timing_char_data.csv`
- STREAM library sweep results: `work/timings.csv`
- Components area: `../README.md`
- Repository guide: `/CLAUDE.md`